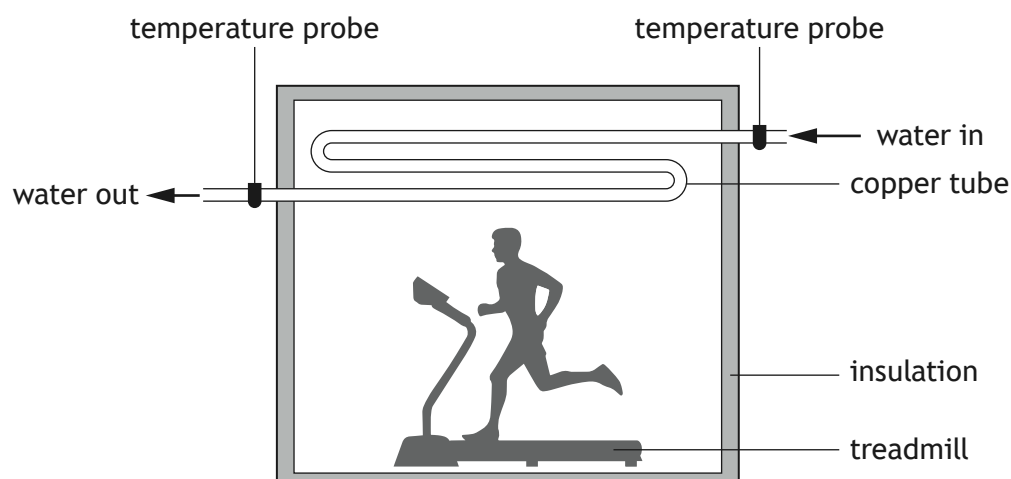


9. An investigation was carried out to study the effect of intensity of exercise on metabolic rate.

A calorimeter was used to determine metabolic rate as shown in the diagram.



An individual walked on a treadmill in the calorimeter for 30 minutes and the temperature increase of the water was calculated every 5 minutes.

The procedure was repeated in another calorimeter with a different individual who ran on the treadmill.

The results are shown in the table.

Time (minutes)	Temperature increase of the water ($^{\circ}\text{C}$)	
	Walking	Running
0	0	0
5	0.2	0.4
10	0.3	0.6
15	0.5	0.9
20	0.8	1.1
25	1.1	1.3
30	1.4	1.6

- (a) Explain how the design of the calorimeter allowed metabolic rate to be determined.

2

- (b) Name the independent variable in this investigation.

1



9. (continued)

- (c) Identify two variables, not already mentioned, that should be controlled for a valid conclusion to be drawn.

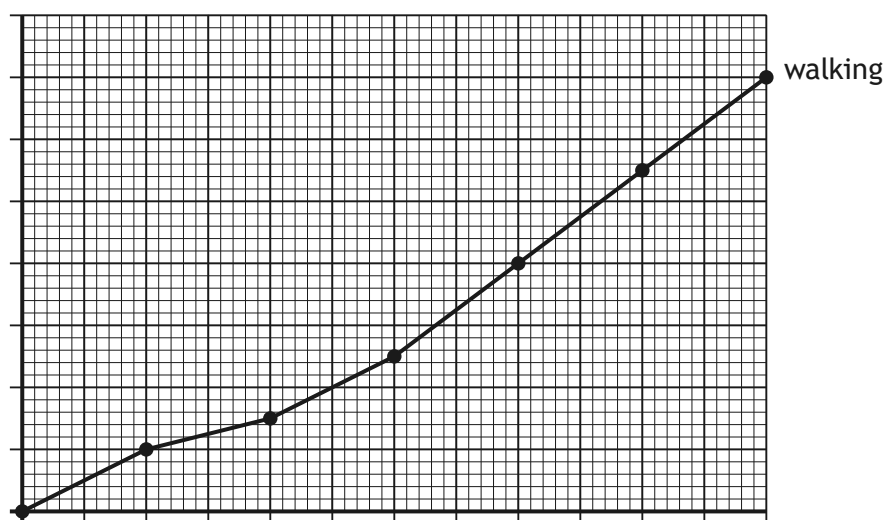
2

1. _____

2. _____

- (d) (i) On the grid, complete the line graph to show the results for **running**.
(Additional graph paper, if required, can be found on *page 31.*)

2



- (ii) Predict the temperature increase after 35 minutes of walking on the treadmill.

1

_____ °C

- (e) Draw a conclusion from the results of this investigation.

1

[Turn over